

AI-Driven Supplier *Risk Intelligence System*

Supply Chain Analytics

Quality Engineering • AI • Predictive Risk

OVERVIEW

Introduction — Why Supplier Risk?

Risk Factor

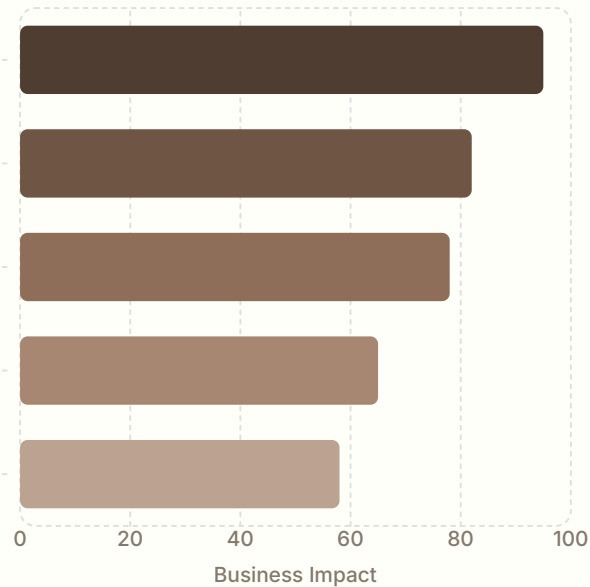
Production Shutdowns

Quality Defects

Delivery Failures

Contract Penalties

Cost Overruns



The Supply Chain Imperative

Modern manufacturing depends on global supply chains. A single supplier failure can trigger cascading consequences across operations, quality, and customer satisfaction.

- Production shutdowns & missed schedules
- Quality defects & contract penalties
- Increased manufacturing costs
- Customer dissatisfaction

For aerospace, defense, semiconductor, and healthcare industries, supplier reliability is directly linked to national security, public safety, and operational readiness.

Why This Project Matters in the AI Era

From Reactive to Predictive

Traditionally, supplier evaluations occur *after* problems arise — quarterly or annual scorecards reviewed too late. AI enables continuous supplier monitoring that fundamentally changes the question organizations ask.

Old Question

"What happened with our supplier?"

New Question

"Which supplier is likely to become a problem next month?"



Forecast Delivery Delays

Predict shipment failures before they disrupt production schedules.



Predict Quality Issues

Identify suppliers trending toward defects before product is received.



Estimate Financial Risk

Quantify cost exposure and recommend alternate sourcing strategies.



Improve Procurement

Empower buyers with data-driven supplier recommendations.

GOALS

Project Objectives

Risk Prediction

Predict supplier risk scores and detect performance deterioration before it impacts operations.

Performance Monitoring

Analyze supplier delivery performance, monitor quality, and evaluate costs across the supply base.

Strategic Sourcing

Improve sourcing decisions, support supplier development programs, and reduce supply chain disruptions.



Supplier Metrics & Variables

Input Variables

Supplier Information Supplier ID · Location · Product Category · Contract Value	Delivery Metrics On-Time Delivery · Lead Time · Late Deliveries · Shipment Accuracy
Quality Metrics Defect Rate · NCR Count · PPM · First Pass Yield	Cost & Operational Purchase Cost · Warranty Cost · Capacity Utilization · Audit Score

Output Variables

Supplier Risk Score Overall composite risk rating per supplier.	Performance Rating Weighted score across quality, delivery, cost, and audit.
Predicted Delays Probability of delivery delay in the next period.	Overall Ranking Supplier reliability index and procurement risk score.

Mathematical Concepts

Defect Rate

$$\text{Defect Rate} = \frac{\text{Defective Units}}{\text{Total Units}}$$

Measures supplier quality performance.

On-Time Delivery

$$\text{OTD} = \frac{\text{On-Time Deliveries}}{\text{Total Deliveries}} \times 100$$

Measures supplier delivery reliability.

Supplier Performance Score

$$\text{Performance} = (0.35 \times Q) + (0.30 \times D) + (0.20 \times C) + (0.15 \times A)$$

Weighted score combining quality, delivery, cost, and audit dimensions. Weights can be adjusted based on organizational priorities.

Risk Score

$$\text{Risk} = w_1Q + w_2D + w_3C + w_4F$$

Where Q = Quality Risk, D = Delivery Risk, C = Cost Risk, F = Financial Risk.

Z-Score (Anomaly Detection)

$$Z = \frac{x - \mu}{\sigma}$$

Detect suppliers performing outside normal variation.

Pareto Analysis

80% of supplier problems originate from **20%** of suppliers. Control Charts (P Charts, X-bar, Moving Range) monitor supplier stability over time.

System Design & Workflow



Each stage of the pipeline builds on the last — from raw ERP data ingestion through statistical analysis, machine learning inference, and finally actionable executive recommendations.

Technologies Used



Programming

- Python
- SQL



Analytics

- Pandas
- NumPy
- Scikit-Learn



Visualization

- Power BI
- Tableau
- Matplotlib



Supply Chain

- SAP / Oracle
- ERP Data
- Supplier Portals



Quality Standards

- APQP · PPAP · FMEA
- AS9100 · ISO 9001
- Lean Six Sigma

ML Models & Algorithms



Regression

- Linear Regression
- Random Forest Regression



Classification

- Random Forest
- XGBoost
- Logistic Regression



Anomaly Detection

- Isolation Forest



Prediction Targets

- Supplier Risk Score
- Delivery Delay Probability
- Quality Failure Probability
- Procurement Risk

Code Example — Random Forest Classifier

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

df = pd.read_csv("supplier_data.csv")

features = [
    "delivery_rate",
    "defect_rate",
    "cost_variance",
    "audit_score",
    "lead_time"
]

target = "risk_level"

X = df[features]
y = df[target]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)

predictions = model.predict(X_test)

print("Supplier Risk Model Complete")
```

How It Works

1

Load Data

Supplier performance data ingested from ERP or CSV exports.

2

Select Features

Five key KPIs: delivery rate, defect rate, cost variance, audit score, lead time.

3

Train Model

75/25 train-test split; Random Forest learns risk patterns.

4

Predict Risk

Classify suppliers into Low, Medium, or High risk categories.

Key Performance Indicators

Track supplier health across all critical dimensions.

KPIs feed directly into the risk score model and executive dashboards.

KPI	Meaning
Supplier Risk Score	Overall supplier risk composite
On-Time Delivery	Delivery performance percentage
Defect Rate	Product quality measure
Supplier PPM	Parts per million defects
Audit Score	Compliance level rating
Lead Time	Delivery responsiveness
Cost Variance	Financial performance vs. budget
First Pass Yield	Manufacturing quality efficiency
Supplier Reliability	Overall consistency index

INDUSTRY APPLICATIONS

Who Benefits Most

Aerospace

Aircraft manufacturing · Engine production · Avionics suppliers

Semiconductor

Wafer suppliers · Chemical suppliers · Equipment vendors

Cloud Infra

Server suppliers · Electrical equipment · Cooling systems

Defense

Missile systems · Defense electronics · Military logistics · Mission assurance

Automotive

Tier 1 & Tier 2 suppliers across complex OEM supply chains

Medical

Sterile components · Electronic assemblies · Regulated supply chains

Key industry partners include Boeing, RTX, Leidos, Northrop Grumman, Lockheed Martin, General Dynamics, Honeywell Aerospace, Collins Aerospace, Intel, TSMC, Samsung Semiconductor, AWS, Microsoft Azure, and Google Cloud.

Before vs. After AI Supplier Intelligence

Without Supplier Intelligence

Late deliveries & production shutdowns

Poor quality & contract penalties

Increased inventory carrying costs

Customer dissatisfaction & lost contracts

With AI Supplier Intelligence

Better Selection

Data-driven supplier qualification and sourcing decisions.

Reduced Disruptions

Proactive risk mitigation before failures occur.

Lower Costs

Reduced procurement, warranty, and penalty costs.

Higher Quality

Suppliers continuously monitored against quality thresholds.

SKILLS DEMONSTRATED

Competency Framework



Supply Chain Analytics

- Supplier Scorecards
- Procurement Analytics
- Risk Assessment



Quality Engineering

- APQP · PPAP · FMEA
- SPC
- Root Cause Analysis



Data Science

- Machine Learning
- Classification Models
- Predictive Analytics
- Dashboard Development



Business Analytics

- KPI Reporting
- Executive Dashboards
- Financial Analysis
- Decision Support

ROADMAP

Future Improvements

ERP Integration Real-time SAP supplier dashboard with live data feeds.	Digital Twin Supplier Digital Twin for simulation and scenario analysis.
ESG Scoring Supplier sustainability scoring aligned with ESG frameworks.	Geopolitical Risk Geopolitical and weather/logistics disruption prediction models.
AI Contract Risk Generative AI for automated contract risk analysis and review.	Gen AI Assistant Generative AI procurement assistant for buyer decision support.

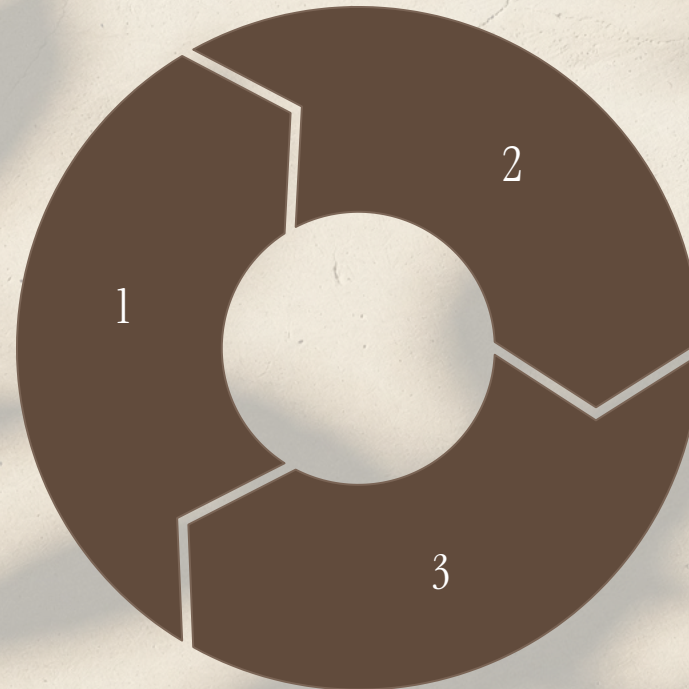
Each enhancement extends the platform's capability from reactive reporting toward a fully autonomous supplier intelligence ecosystem.

FINAL STATEMENT

Supplier Risk Intelligence

This project demonstrates how supply chain analytics, quality engineering, business intelligence, and artificial intelligence can work together to identify supplier risks before they disrupt operations. In today's AI-driven industrial landscape, organizations that can predict supplier performance rather than simply react to supplier failures gain a significant advantage in cost, quality, resilience, and mission success.

Predict
Use AI to forecast supplier risk before disruptions occur.



Monitor

Continuously track KPIs across the entire supply base.

Act

Empower procurement with data-driven sourcing decisions.